DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Winter Examination – 2022

Course: B. Tech. **Branch : Electrical Engineering** **Semester : VII**

Subject Code & Name: High voltage Engineering (BTEEC702)

Max Marks: 60 **Date: 30/01/2023** **Duration: 3 Hr.**

Instructions to the Students:

1. *All the questions are compulsory.*
2. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.*
3. *Use of non-programmable scientific calculators is allowed.*
4. *Assume suitable data wherever necessary and mention it clearly.*

		(Level)	Marks
Q. 1	**Solve Any Two of the following.**		**12**
A)	Derive Poisson's Equation. Also Write the Laplace's equation.	Evaluation	6
B)	Explain Estimation and Control of Electric Stress.	Understand	6
C)	Define the terms: (a) Disruptive Discharge Voltage (b) Withstand Voltage (c) Fifty Percent Flashover Voltage (d) Hundred Percent Flashover Voltage (e) Creepage Distance (f) B.C. Test Voltages	Remember	6
Q.2	**Solve Any Two of the following.**		**12**
A)	Explain Ionization by Collision and Photo-ionization.	Understand	6
B)	Write a short note on Time lags for Breakdown.	Remember	6
C)	Explain the Streamer theory of breakdown in air at atmospheric pressure.	Understand	6
Q. 3	**Solve Any Two of the following.**		**12**
A)	With neat Sketch explain Liquid purification system with test cell in case of Pure and commercial liquids.	Understand	6
B)	Explain i) Suspended Particle Mechanism ii) Cavitation and Bubble Mechanism	Understand	6
C)	Prove that $\frac{V}{d_0} = E_a = 0.6 \left[\frac{Y}{\epsilon_0 \epsilon_r}\right]^{\frac{1}{2}}$ In case of Electromechanical Breakdown	Evaluation	6
Q.4	**Solve Any Two of the following.**		**12**
A)	Explain with suitable figures the principles and functioning of (a) expulsion gaps and (b) protector tubes.	Understand	6
B)	What is a surge diverter? Explain its function as a shunt protective device.	Remember	6
C)	Explain Power Frequency Tests, Impulse Voltage Tests and Thermal Tests of Bushings.	Understand	6

Q. 5	Solve Any Two of the following.		12
A)	Explain with neat circuit diagram and waveform the generation of High DC Voltages using Half wave and Full Wave rectifier circuit.	Understand	6
B)	Explain with circuit diagram the generation of High AC Voltages using Cascade transformer.	Understand	6
C)	Discuss Measurement of High Direct Current Voltages using high ohmic series resistance with Microammeter and Resistance Potential Divider.	Create	6

*** End ***